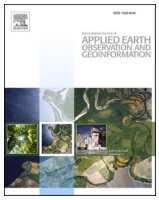
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A multi-source remote sensing-based geocommunication tool for global flood monitoring and management

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ABSTRACT

Global warming is expected to increase the frequency of extreme flooding, making rapid and accurate flood mapping crucial for effective risk assessment. Many governmental agencies and organizations are developing flood risk assessment tools; however, due to the lack of observational records, some rely on probabilistic generated data, uncalibrated simulations, or terrain-based methods, all of which are subject to various types of uncertainty. Although remote sensing provides valuable flood data with global coverage, single-source reliance is constrained by satellite revisit rates, resolution, weather conditions, and sensor limitations. To address these challenges, this study introduces a user-friendly application on the Google Earth Engine (GEE) platform that enables near-real-time global flood mapping using a multi-source remote sensing approach. By leveraging optical and SAR imagery, the App ensures improved water detection accuracy and supports all-weather and day/night monitoring. Our results show SAR and optical flood inundation maps agree up to 80 %. Beyond flood mapping, the tool leverages GEE datasets to extract multi-disciplinary information, such as population exposure and affected residential, urban, and cropland areas, to support timely decision-making. For example, during the Sylhet, Bangladesh flood, the tool identified over 300,000 people potentially affected and approximately 600 km^2 of cropland inundated. This research presents one of the first global-scale, rapid, multi-source flood mapping tools tailored to both expert users and non-expert decision-makers. It offers a practical solution to current data limitations and supports more informed emergency response, planning, and climate resilience efforts to foster communication across scientific, policy, management, and operational communities.

1. Introduction

Flooding represents one of the most frequent and destructive natural hazards globally, with recurring impacts on human life, infrastructure, and economic systems (Jafarzadegan et al., 2023; Alipour et al., 2020; Samadi et al., 2025; Ahmadalipour and Moradkhani, 2019). These events can result from a combination of natural processes and anthropogenic influences, including intense precipitation, riverine overflow, snowmelt, dam failure, and inadequate land-use planning (Abegaz et al., 2024). Although floods have historically posed significant risks, past research highlights the role of climate change in intensifying key flood-inducing factors, such as extreme rainfall and sea-level rise over recent decades (Hamidi et al., 2025a; Davenport et al., 2021; Taherkhani et al., 2020; Halmstad et al., 2013). This trend is projected to persist, contributing to the spatial and temporal expansion of flood-prone areas in the future (Johnson et al., 2019). In parallel, population growth and accelerating socioeconomic development are driving increased urbanization within flood-susceptible regions, thereby elevating exposure and the potential for significant damage and loss of life in future flood scenarios (Hamidi et al., 2025b; Yarveysi et al., 2025; Tripathy et al., 2024; Jafarzadegan et al., 2023; Yarveysi et al., 2023; Zhang et al., 2018). Between 2000 and 2019, flooding caused global damages of approximately USD 651 billion, affecting 1.6 billion people (Devitt et al., 2023). Thus, the need for a comprehensive approach that accurately quantifies the three elements of risk—hazard, exposure, and vulnerability is invaluable (Yarveysi et al., 2025; Albano et al., 2020). This approach is necessary for identifying exposed areas, developing appropriate flood management strategies, assisting in decision-making for preparedness, response, and recovery, and ultimately promoting risk-informed resilience planning. Accordingly, there is a growing emphasis on developing

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strategies ranging from regional to continental scales to mitigate the impacts of this persistent and potent natural phenomenon (Dottori et al., 2016).

Developed countries have collected abundant and high-quality data and developed tools to enhance the resilience of communities to extreme flood events and devise efficient flood risk assessment and risk management strategies (Table 1). These nations have developed their flood risk assessment approaches tailored to their specific geographic areas (e.g., the HAZUS M−H Flood model in the United States, HISS-SSM in the Netherlands and FLEMO in Germany) (Glas et al., 2019). For instance, the European Centre for Medium-Range Weather Forecasts (ECMWF) offers the Global Flood Awareness System (GloFAS) to aid in flood preparedness globally, especially for large transnational river basins. Moreover, the European Flood Awareness System (EFAS) forecasts large-scale riverine and flash floods, calculating both hazard and impact predictions. The Natural Sciences and Engineering Research Council of Canada (NSERC) and the University of Quebec offer the Flood Map Viewer for visualizing floodplain changes in Canada due to climate change. The United Nations University's Flood mapping tool (FMT) uses open Earth data for flood risk management, including early warning systems. The Global Flood Partnership (GFP) is a community of scientists and organizations that mobilize to provide data and map-based support in response to emergent flood events. The United States (U.S.) coastline, particularly along the Atlantic and Gulf coasts, is highly vulnerable to flooding from heavy rains and tropical storms, prompting significant investments in flood assessment and protection, including the National Hurricane Center's Storm Surge Risk Maps, FEMA's Flood Map Service Center and the National Flood Hazard Layer, and The U.S. Geological Survey (USGS)'s Flood Inundation Mapping tool and Flood Event Viewer. Additionally, the National Aeronautics and Space Administration (NASA)'s Flooding Analysis Tool assesses sea level rise impacts. The U.S. National Oceanic and Atmospheric Administration (NOAA) offers tools like the Sea Level Rise Viewer, Coastal Flood Exposure Mapper, and Coastal Inundation Dashboard for flood data. While some of these advancements provide local communities with a detailed flood maps and impact estimates, there remains a pressing need for a comprehensive, globally accessible tool. Despite efforts to create global flood risk models, data scarcity and limited cyberinfrastructure capacity continue to pose significant barriers to conducting flood risk assessments, especially in developing nations (Glas et al., 2019; UNDRR, 2019). The advancement of modern remote sensing technologies integrated within cloud-based platforms, alongside geospatial, demographic, and socioeconomic datasets, presents a significant opportunity to address this gap and enhance the effectiveness of global flood monitoring and impact assessment (Hamidi et al., 2023).

Table 1
Overview of established flood mapping tools and platforms.

Flood Tool/(Provider)	Map Type	Data Sources	Coverage
Global Flood Awareness System (GloFAS)/(Copernicus)	Statistical	ECMWF forecasts, LISFLOOD, Sentinel-1 SAR	Global
European Flood Awareness System (EFAS)/(Copernicus)	Statistical	ECMWF forecasts, LISFLOOD, Sentinel-1 SAR	Europe
Flood Map Viewer/(NSERC & Univ. of Quebec)	Statistical	Remote sensing, hydrological models	Canada
Flood Mapping Tool/(United Nations University)	Historical data	Landsat optical imagery	Global
Global Flood Partnership/(Global Flood Partnership)	Statistical	MODIS, Sentinel, Landsat, Hydrologic Models	Global
Storm Surge Risk Maps/(National Hurricane Center)	Statistical	Storm surge models (SLOSH), tide data	USA (coastal)
Flood Map Service Center/(FEMA)	Statistical	FEMA hydraulic models, DEMs	USA
National Flood Hazard Layer (NFHL)/(FEMA)	Statistical	FEMA Flood Insurance Rate Maps, LiDAR DEMs	USA
Flood Inundation Mapping Tool/(USGS)	DEM based	USGS streamflow, LiDAR, HEC-RAS	USA
Flood Event Viewer/(USGS)	Historical data	USGS high-water marks, sensors	USA
Flooding Analysis Tool/(NASA)	Statistical	NOAA tide gauges, statistical models	USA (coastal)
Sea Level Rise Viewer/(NOAA)	DEM based	LiDAR DEMs, NOAA scenarios	USA (coastal)
Coastal Flood Exposure Mapper/(NOAA)	Statistical	FEMA maps, NLCD, DEM, census data	USA (coastal)
Coastal Inundation Dashboard/(NOAA)	DEM based	Tide gauge data, LiDAR DEMs	USA (coastal)

Over the past decades, notable advancements in satellite technology have led to its effective application across various fields, notably for observing land dynamics and detecting large-scale changes to forests and water (Mahdipour et al., 2024; Moradi and Sharifi, 2023; Sogno et al., 2022). Specifically, satellite products have enabled data-scarce regions to leverage remote sensing for flood hazard mitigation, helping to fill critical gaps caused by limited data availability (Glas et al., 2019; Munasinghe et al., 2018). Optical and Synthetic Aperture Radar (SAR) satellite imagery are valuable sources for flood monitoring; however, they operate differently and possess unique characteristics, each offering distinct strengths and limitations. For example, Optical satellites, such as Landsat, have been in operation since the 1970 s and provide valuable multispectral imagery for flood mapping. These sensors detect floodwaters by capturing reflected solar radiation in the visible, infrared, and shortwave infrared wavelengths (Marghany, 2022a; Lin et al., 2016). However, they rely on clear weather conditions i.e., they cannot penetrate cloud cover and are ineffective at night due to their dependence on sunlight. These limitations make optical imagery less reliable for flood monitoring, particularly during storms or heavy rainfall when clouds are abundant (Hamidi et al., 2023; Grimaldi et al., 2020).

Alternatively, modern SAR satellites, such as Sentinel-1, use microwave radar signals, allowing them to capture flood imagery regardless of weather conditions or time of day (Marghany, 2022b). Unlike optical sensors, SAR transmits radar pulses and records their reflections from the Earth's surface. Smooth water surfaces typically reflect signals away from the sensor, appearing dark in SAR images (Marghany, 2024). This capability enables surface water observations under all weather and lighting conditions (Zhang et al., 2022). However, SAR also presents challenges: rough water surfaces caused by wind or debris may appear brighter, and highly reflective surfaces, such as wet soil, can sometimes be misclassified as water (Notti et al., 2018). It is worth noting that SAR can detect water beneath vegetation, making it highly effective in forested or densely vegetated areas, particularly when using longer wavelengths such as L-band (Marghany, 2024).

Single-source remote sensing data have limitations, including satellite revisit rates, spatial resolution, and sensor defects, which can impede timely flood mapping (Hamidi et al., 2023). Multi-source methods can overcome some of these limitations by leveraging the strengths of each remote sensing data product (Tong et al., 2018). For instance, combining optical and SAR data enables accurate water extraction with color-rich optical imagery under clear skies, while leveraging SAR's all-weather and day/night capability to penetrate cloud cover during flood events (Marghany, 2024; Tong et al., 2018). Thus, by leveraging both technologies, we can take advantage of their distinct strengths to achieve a more reliable and comprehensive assessment for flood monitoring. Furthermore, integrating both optical and SAR data can enhance advanced machine learning models and support time-series analysis (Drakonakis et al., 2022). Accordingly, this research aims to utilize multi-source remote sensing data to support the growing need for reliable flood-risk management practices and decision-support tools (Shan et al., 2022).

This study introduces a novel flood mapping tool that leverages

optical and SAR remote sensing techniques with geospatial analysis in a cloud-based environment. We extend on previous multi-source remote sensing approaches for flood extent mapping (Hamidi et al. (2023) and develop an advanced, automatic, large-scale, near-real-time flood inundation mapping tool with global coverage. This tool utilizes the capabilities of the Google Earth Engine (GEE) platform, leveraging multiple datasets and cloud computing resources to provide decision-makers and responsible authorities with critical flood-related information. This product is intended to assist in strategic flood response planning, global flood hazard observation, and vulnerability estimation. Additionally, the tool provides experts with options for detailed analysis of the final information. Its open-access design further enhances transparency and reproducibility, while offering a flexible framework for future development and broader adoption within the scientific and disaster management communities.

This paper is organized as follows. Section 2 presents the materials and methods used in this study, including the datasets (2.1), study areas (2.2), methodology (2.3), evaluation formulation (2.4), and socioeconomic exposure estimation (2.5). Section 3 presents the results, focusing on four extreme flood events in diverse regions: Coffeyville, Kansas, U.S. (3.1); Tetouan, Morocco (3.2); Saga, Japan (3.3); and Sylhet Division, Bangladesh (3.4). Section 4 provides a discussion of the findings, while Section 5 concludes the study by summarizing key contributions and outlining directions for future work.

2. Materials and methods

2.1. Datasets

The Sentinel-1 a European Space Agency (ESA) radar mission, launched in October 2014, provides dual-polarization C-band SAR data with a 12-day revisit rate, three spatial resolutions (10, 25, and 40 m), and four band combinations (VV, HH, VV + VH, HH + HV). It operates in four SAR modes: stripmap, interferometric wide (IW) swath, extra wide (EW) swath, and wave (WV) (Malenovský et al., 2012). Sentinel-2, also an ESA satellite, offers multispectral imagery with 13 spectral bands, a 290 km swath width, and 10–60 m spatial resolution, depending on the band. This study utilizes 10-m Sentinel-1 Level-1 GRD ("COPERNICUS/S1_GRD") data with Vertical-Horizontal polarization (VH), ascending pass direction, and Interferometric Wide (IW) mode. Additionally, it employs 20-m Sentinel-2 ("COPERNICUS/S2″) imagery in Google Earth Engine as an optical data source for global flood detection and monitoring. The 30-meter NASADEM digital elevation data (NASA JPL, 2020) and the European Commission's Joint Research Centre (JRC) Global Surface Water Mapping Layers (Pekel et al., 2016) are used to filter out areas with slopes greater than 5° and help exclude permanent water bodies, respectively. The application also incorporates Land Cover Land Use (LCLU) layers, such as the ESA WorldCover product (Zanaga et al., 2022) and the USGS National Land Cover Dataset (NLCD) (Dewitz, 2023), to evaluate the impact on cropland and urban areas. Additionally, the application incorporates WorldPop Global Project Population Data (Sorichetta et al., 2015) to estimate the number of people at risk from floods and uses Local Climate Zones (LCZs) (Demuzere et al., 2022) to assess the extent of flood damage to different urban and rural landscape types (Hamidi and Peter, 2024). Table 2 provides a summary of the data sets used in this study.

Table 2
The summary of datasets used in the GEE App.

Data	GEE ID	Spatial Resolution
Sentinel-1	"COPERNICUS/S1_GRD"	10 *m*
Sentinel-2	"COPERNICUS/S2″	20 *m*
LCLU for the Globe	"ESA/WorldCover/v200″	10 *m*
LCLU for the USA	"USGS/NLCD_RELEASES/2019_REL/NLCD"	30 *m*
DEM	"NASA/NASADEM_HGT/001″	30 *m*
World Population	"WorldPop/GP/100 m/pop"	100 *m*
Local Climate Zones	"RUB/RUBCLIM/LCZ/global_lcz_map/latest"	100 *m*
Surface Water Mapping	"JRC/GSW1_0/GlobalSurfaceWater''	30 *m*

2.2. Study areas

The strength of this App lies in its fast and efficient global flood extent mapping capabilities. Users can perform flood inundation analysis without the need for data preparation or extensive computational resources and can simply define the flood event period and draw a polygon around the target study area. To demonstrate the App's applicability, we applied it to four extreme flood events from around the globe: Coffeyville, Kansas, U.S. (May 2019); Tetouan, Morocco (January 2021); Saga, Japan (August 2021); and Sylhet Division, Bangladesh (June 2024). Fig. 1 shows the locations of these case studies, along with other examples presented in the App's report, which is available in the supplementary materials.

On May 20, 2019, an intense low-pressure system brought heavy rain and persistent flooding to Kansas, U.S., causing all rivers in central and south-central areas to flood, with some reaching levels not seen in over a decade. This extreme flash flooding that led to evacuations in southeast Kansas due to severe inundation (Francis and Melliger, 2021). Moreover, in this study, we assess flood risk in Tetouan, Morocco, where urban expansion and increased impervious surfaces exacerbate runoff, overload drainage systems, and heighten flood vulnerability. The extreme flood in early January and March 2021 had significant impacts in Tetouan, with approximately 275 houses waterlogged, 11 cars swept away, some facility exteriors collapsing, and several roads rendered completely impassable, while others experienced prolonged traffic disruptions (El Yacoubi et al., 2023). Additionally, in mid-June 2024, heavy rains caused severe flooding in Sylhet Division, Bangladesh, already affected by earlier floods from late May. This flood event affected 4.6 million people and caused 16 deaths (Doomun et al., 2024). Another extreme event in August 2021 due to a slow-moving and strengthening front brought torrential rain to Japan, severely impacting areas such as Saga, Fukuoka, and Hiroshima. This event led to 13 deaths, the complete destruction of approximately 50 houses, and partial destruction of around 2,000 houses (Shao et al., 2022; Synspective, 2021). We use these diverse extreme flood events from around the world to demonstrate the effectiveness of our tool for global flood inundation mapping.

2.3. Methodology

This App leverages multi-source remote sensing datasets to enable precise and efficient floodwater extraction, following the approach suggested by Hamidi et al. (2023). The App automates the entire process and incorporates additional globally available datasets to provide more detailed information in a user-friendly format. Leveraging the GEE cloud platform, the App eliminates the need for users to download data or invest in local computational resources. Users simply enter the flood event period and draw a boundary around the study area in the App's environment (see the required Steps 1 and 3 in Fig. 2). The App then automatically collects SAR images from before and after the flood and calculates the Normalized Difference Flood Index (NDFI) and the Difference Image Index (DII), which are formulated as (Hamidi et al., 2023):

$$NDFI = \frac{|mean(\sigma_{0_{[BF]}})| - |min(\sigma_{0_{[AF]}})|}{|mean(\sigma_{0_{[BF]}})| + |min(\sigma_{0_{[AF]}})|} \quad (1)$$

$$DII = |min(\sigma_{0_{[AF]}})| - |mean(\sigma_{0_{[BF]}})| \quad (2)$$

where $\sigma_{0[BF]}$ and $\sigma_{0[AF]}$ are the backscatter coefficient of SAR imagery

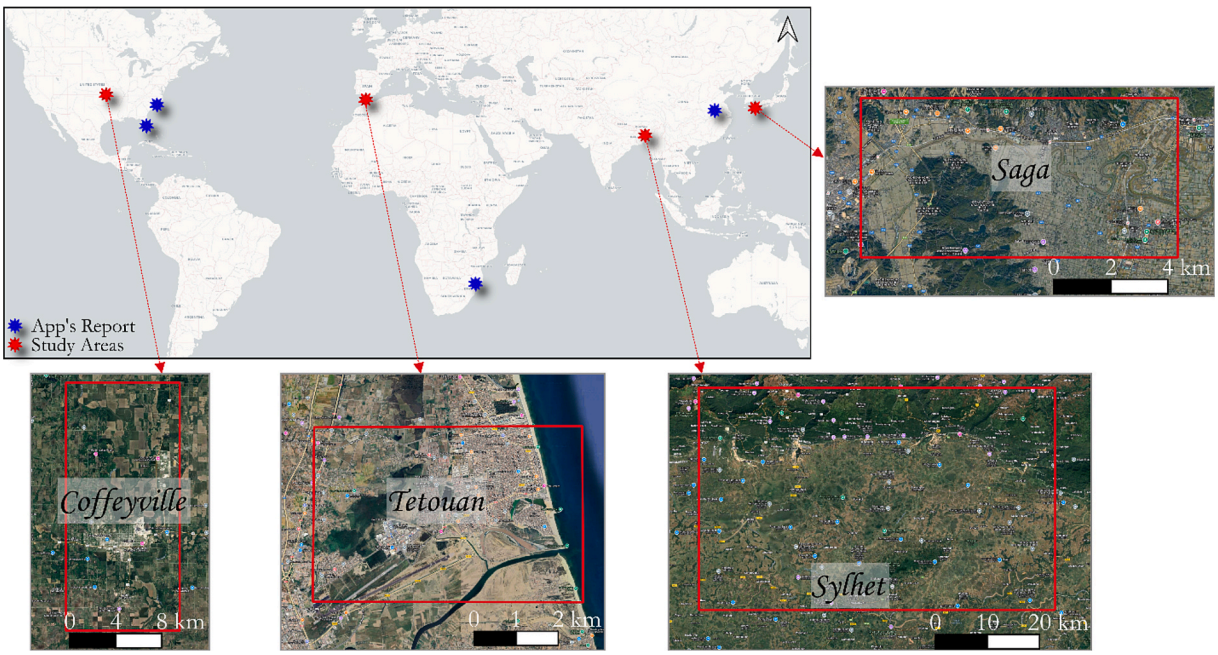


Fig. 1. Study areas used in this research. Red stars indicate case studies presented in this paper; blue stars represent additional cases from the App's report in the supplementary materials. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

before and after flood, respectively. For extracting flooded pixels and generating the flood map, a threshold formulation based on Equation (3) is used; a minus sign was applied for NDFI (below threshold) and a plus sign for DII (above threshold) to identify flooded pixels.

$$Thr = mean(SAR\ Index\ Image) \pm k_f \times stdev(SAR\ Index\ Image) \quad (3)$$

where the k_f coefficient is set at 1.5 in similar research (Cian et al., 2018; Clement et al., 2018). This tool uses the flood inundation map extracted with this constant threshold value as the primary estimate of flood extent; however, the accuracy of this map is not guaranteed (Hamidi et al., 2023). To improve accuracy, the App includes an automated sensitivity analysis algorithm that identifies optimal threshold values using flood maps derived from available optical data. This feature can be activated via an optional checkbox embedded in the App interface (Fig. 2, "2. Advanced Flood Map Features", Calculate Optimum Thresholds) that allows users to incorporate optical datasets for SAR threshold calibration, thereby improving the generated flood inundation map. By selecting this checkbox, the App utilizes available Sentinel-2 optical images (if not entirely affected by cloud cover) to calculate the Modified Normalized Difference Water Index (MNDWI),

$$MNDWI = \frac{Green\ Band - SWIR\ Band}{Green\ Band + SWIR\ Band} \quad (4)$$

where the *Green Band* is Band 3 and *SWIR Band* (Short Wave InfraRed Band) is Band 11 of Sentinel-2.

2.4. Evaluation formulation

To find the optimal SAR index threshold values, the App performs an automated iterative sensitivity analysis by testing different k_f values in Equation (3). It then evaluates the resulting SAR and optical flood maps using the Total Agreement (*ToAgr*) equation (Hamidi et al., 2023),

$$ToAgr\ \% = (SAR \cap Optical) / (SAR \cup Optical) \times 100 \quad (5)$$

where the intersection of the two flood maps is divided by their union. In other words, it measures the area classified as flooded by both datasets divided by the total area classified as flooded by either SAR or optical imagery. Equation (5) serves as an effective criterion for identifying the optimal SAR threshold values referenced in Equation (3). Lower threshold values result in a reduced spatial overlap with the optical flood inundation map, thereby minimizing the numerator in Equation (5). Conversely, higher threshold values increase the total mapped area from both SAR and optical sources, leading to a larger denominator. This trade-off between intersection and total flood extent enables the derivation of a distinct threshold that maximizes the consistency between SAR-derived flood extents and the reference optical data flood maps, thereby ensuring an optimal balance between omission and commission errors. Finally, to report the final agreement between SAR and optical flood inundation maps, the App uses the Agreement (*Agr*) formulation (Hamidi et al., 2023),

$$Agr\ \% = (SAR \cap Optical) / Optical \times 100 \quad (6)$$

where the intersection of the two flood maps is divided by optical flood map. Equation (6) quantifies the proportion of the SAR-derived flood extent that overlaps with the optical-derived flood map, treating the latter as the reference. This metric is used to evaluate the accuracy and spatial agreement of the SAR-based flood inundation map relative to the optical benchmark. The ultimate flood map is then derived by merging the final NDFI and DII flood maps, which can be previewed in the GEE map view panel.

2.5. Socioeconomic exposure estimation

Additionally, another optional checkbox (Fig. 2, "2. Advanced Flood Map Features", Extract Strategic Information) allows users to extract multidisciplinary information from the extracted flood maps, highlighting exposed populations and strategic regions, such as residential, cropland, and industrial areas. The App's methodology flowchart is shown in Fig. 3.

The App is designed for rapid flood inundation mapping with minimal user input, making it suitable for non-expert users. To expedite the process, it assumes a constant k_f value of 1.5 for calculating SAR index thresholds, a common value in flood extent estimation (Cian et al., 2018; Clement et al., 2018; Long et al., 2014). Previous studies, such as Long et al. (2014), proposed a fixed threshold value to differentiate flooded

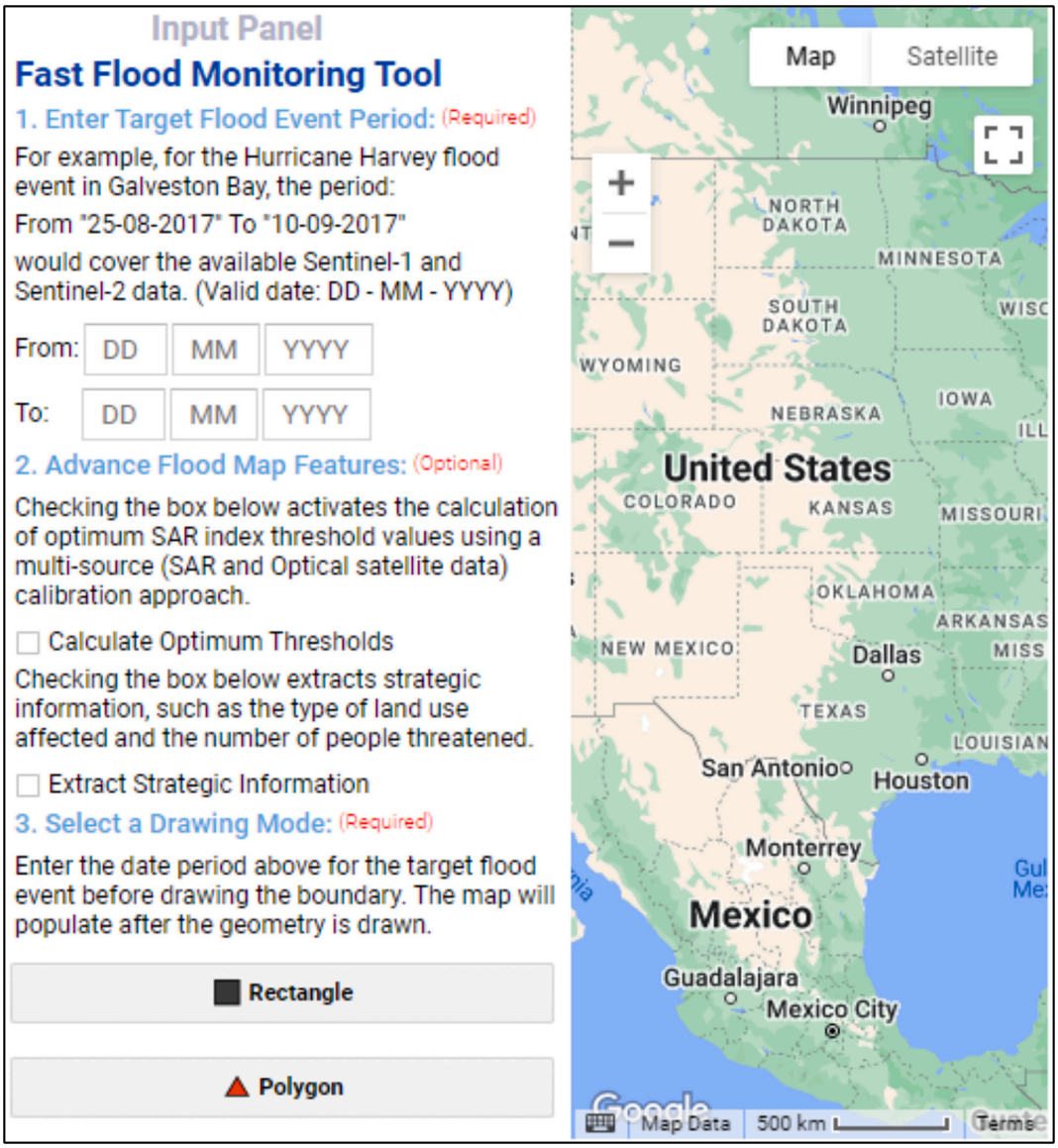


Fig. 2. GEE App interactive page environment.

from non-flooded pixels, a method that has been widely adopted in subsequent flood mapping research (e.g., (Singha et al., 2020; Clement et al., 2018; Cian et al., 2018). In contrast, our approach leverages multi-source remote sensing by combining flood mapping methods based on SAR and optical data, to derive more accurate, event-specific threshold values for SAR-based indices, thereby enhancing the reliability of flood extent delineation. Furthermore, several statistical methods, such as the Bayesian network algorithm (e.g., D'Addabbo et al. (2016)), have been used to determine unsupervised threshold values for SAR imagery. However, these approaches are computationally intensive compared to our multi-source method, which offers a faster and more efficient alternative with comparable accuracy. Furthermore, A constant threshold of 0.4 is used to extract flood pixels from optical index images, consistent with studies like Baig et al. (2013), Phiri et al. (2021), Hamidi et al. (2023), and Ferral et al. (2019). Finally, the reference period is set from "01-06-2017″ to "15-06-2017″, to collect SAR images before the flood event. However, the App may fail to generate a flood extent map if these parameters are not properly defined, such as when no images are available for the preset reference dates, requiring the user to choose an alternative period.

Furthermore, the App provides expert users with additional flexibility to analyze flood maps and obtain detailed information, improving map utility. Features include displaying NDFI and DII images, viewing SAR index pixel value histograms, and using USGS LCLU data to assess higher-quality land cover for the United States. Users must reselect the drawing mode and redraw the boundary after changing parameters or selecting options to update the map. A detailed App's manual is available in the Supplementary materials and in CUAHSI HydroShare resources (Hamidi and Peter, 2024) for further guidance.

3. Results

In this section, we demonstrate the application of this tool for flood extraction in four different extreme flood study cases around the globe: Coffeyville, Kansas, U.S. (May 2019); Tetouan, Morocco (January 2021); Saga, Japan (August 2021); and Sylhet Division, Bangladesh (June 2024).

3.1. Extreme flood in Coffeyville, KS, U.S. (May 2019)

To map the Coffeyville flood, we set the event dates from "20-05-2019″ to "31-05-2019″ and selected options to calculate the optimum

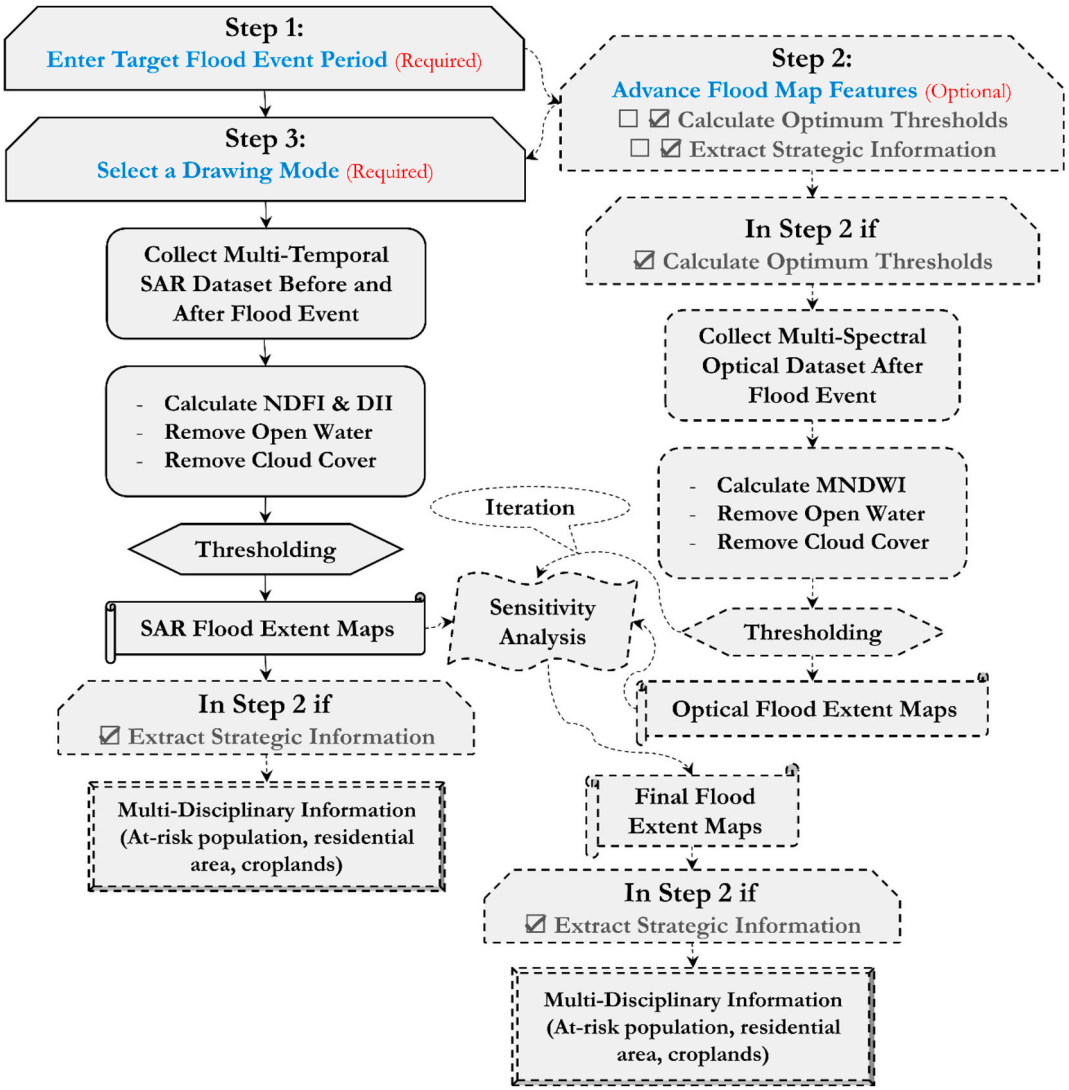


Fig. 3. GEE App methodology flowchart.

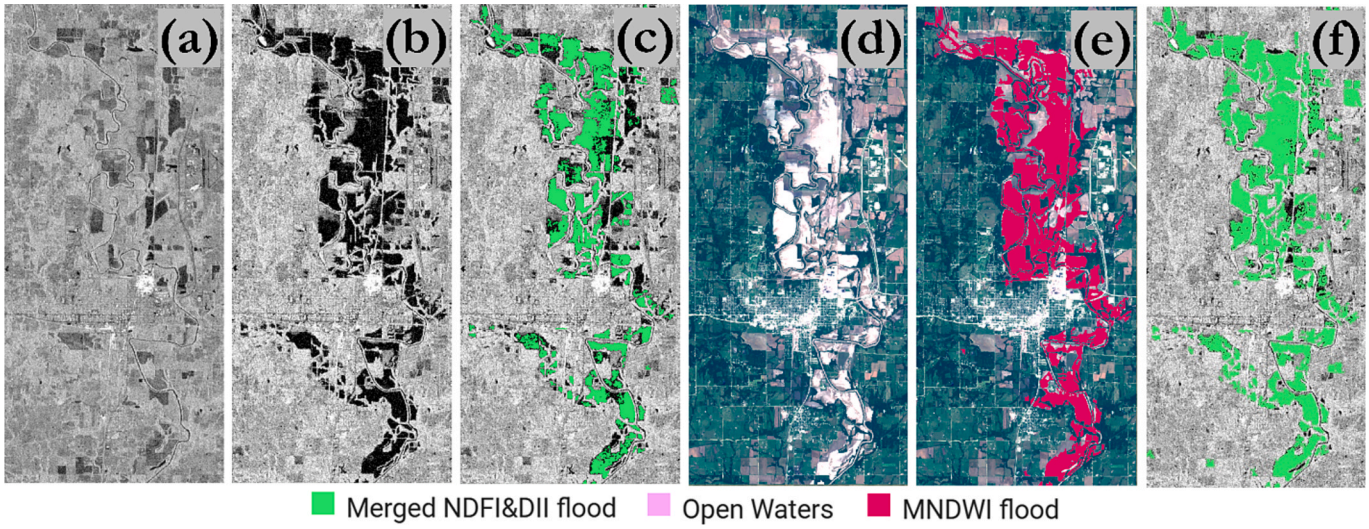


Fig. 4. Coffeyville flood: SAR images (a) before and (b) after the flood and (c) extracted flood areas using the predefined $k_f = 1.0$ (green). (d) Post-flood optical image and (e) extracted flood areas (red). (f) SAR flood map using optimal k_f values calculated from sensitivity analysis ($k_f = 0.6$; refer to Fig. 5). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

threshold and extract strategic information. The results are shown in Fig. 4. Pre- and post-flood SAR images are displayed in Fig. 4a and 4b, with flood areas extracted using a predefined threshold ($k_f = 1.5$) in Fig. 4c. With the "*Calculate Optimum Threshold*" option, the App also uses post-flood optical images (Fig. 4d) to extract flooded pixels (Fig. 4e) via MNDWI (using a 0.4 threshold). The App then performs an automated sensitivity analysis using the optical flood inundation map to optimize the SAR threshold values. Fig. 5 shows that $k_f = 0.6$ is calculated as the optimal threshold values for both NDFI and DII. Table 3 reports that the agreement percentage (*Agr%*) using the optimal SAR threshold values, compared with the optical flood map, is 80 % for NDFI and 79 % for DII. The result map (merging NDFI and DII flood maps using optimum threshold values) is shown in Fig. 4f. Referring to Table 3, the App reports that the majority of the flooded area is cropland, covering 33.32 out of 40.75 square kilometers.

3.2. Extreme flood in Tetouan, Morocco (January 2021)

The extreme flood event in Tetouan occurred in mid-January 2021, and in the App, the flood event date range was set from "10-01-2021″ to "20-01-2021″ with options to calculate the optimum threshold and extract strategic information enabled. Fig. 6a and 6b display the pre- and post-flood SAR images, while the extracted flood areas using a predefined threshold ($k_f = 1.5$) are shown in Fig. 6c. With the "*Calculate Optimum Threshold*" option enabled, the App utilized available optical images (Fig. 6d) to calculate MNWDI, extracting flooded pixels from optical image using a 0.4 threshold value (Fig. 6e). The App then calculates optimal SAR index threshold values by performing an automated sensitivity analysis, comparing the extracted flood areas from optical images. Fig. 7 presents the results of this analysis, with the optimal k_f value calculated as 1.0 for both NDFI and DII. Furthermore, Table 3 reports agreement percentages (*Agr%*) of 67 % and 75 % for NDFI and DII, respectively, when compared with the optical flood map. The result map, merging NDFI and DII flood maps using optimum threshold values, is shown in Fig. 6f. According to Table 3, although the total flooded area was only 3.41 km^2 in Tetouan, the App estimates that 3,293 people were affected, a significant number due to the high population density in the flooded urban area.

3.3. Extreme flood in Saga, Japan (August 2021)

For extracting the August 2021 extreme flood in Saga, Japan, using the proposed App, the flood event start date is set to "15-08-2021″ and the end date to "20-08-2021″. Fig. 8 displays SAR data from before the flood, after the flood, and highlighting the extracted flood areas using predefined SAR indices threshold value ($k_f = 1.5$) in Fig. 8a, 8b and 8c, respectively. Unfortunately, for this case study, all available optical data were completely cloud-covered. As a result, the App was unable to perform a sensitivity analysis to determine optimal threshold values for the SAR indices. Comparing the dark areas of before and after SAR data shows that this predefined k_f value could not extract the entire flooded pixels. While there is no cloud-free optical image available for calibrating k_f values, the App provides an option for users to manually adjust the k_f values for a rough flood estimation and view the updated flood inundation map. For this case study we examined different k_f values and found that $k_f = 0.8$ provides a reasonable estimate for both the NDFI and the DII in extracting flood areas from SAR data, effectively capturing most of changes between pre- and post-flood SAR images. The resulting flood map, generated merging NDFI and DII with $k_f = 0.8$, is shown in Fig. 8d. The extracted multidisciplinary information in Table 3 demonstrates that this region is primarily agricultural with a high population density (7.87 out of 9.77 square kilometers of flooded area is categorized as cropland). Notably, The App's report estimates about 2,161 people were threatened by the time the image was taken in such a small flooded area.

3.4. Extreme flood in Sylhet Division, Bangladesh (June 2024)

It is worth mentioning that, as we prepared this article, the flood in Bangladesh was ongoing, highlighting the tool's advantage for urgent flood inundation mapping. We considered the start date of flood as "15–06-2024" and end date "20–06-2024" to map the flood for a part of the region. The collected before and after flood SAR images are shown in Fig. 9a and 9b, respectively. The optical image for this period is completely cloud-covered, making it unsuitable for SAR threshold calibration. Additionally, the predefined k_f value ($k_f = 1.5$) did not adequately capture the flooded area in the post-flood image. Therefore, by adjusting the k_f values and comparing the pre- and post-flood SAR images, we selected $k_f = 0.5$ to extract the flooded pixels. The resulting flood inundation map is presented in Fig. 9c. Based on the estimated flood, the App reports that nearly 341,553 people were potentially impacted and their lives put at risk (Table 3), for only small part of the Sylhet area. It also indicates that approximately 595.41 out of 749 square kilometers of the flooded area are cropland, which could significantly impact the region's economy and food security. These maps and multidisciplinary data offer responsible authorities valuable insights to

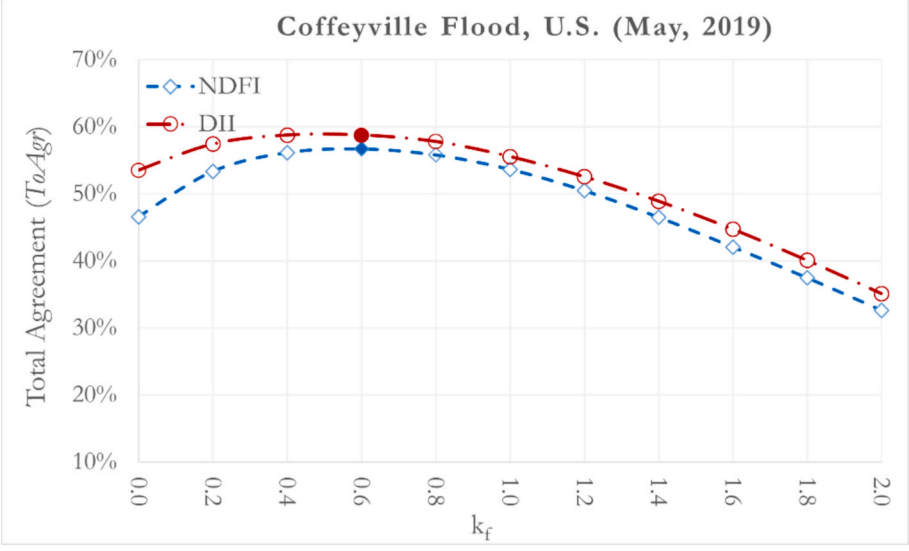


Fig. 5. Sensitivity analysis of NDFI and DII threshold values compared with optical flood inundation map to find optimum k_f values for Coffeyville flood, May 2019.

Table 3
List of flood events assessed using the proposed App in this paper as study areas and the App's manual report which is available in the Supplementary material (Hamidi and Peter, 2024).

EventLocation	Climate Condition	Date	Indices/k_f	Agr (%)	People (No.)	Flood (km^2)			Details
						Total	Urban	Cropland	
Coffeeville, KS, USA	Subtropical	May 2019	NDFI/0.6 DII/0.6	80 % 79 %	5	40.75	0.312	33.32	Study Area
Sylhet, Bangladesh	Tropical	Jun. 2024	NDFI/0.5 DII/0.5	–– ––	341,553	749	0.024	595.41	Study Area
Tetouan, Morocco	Mediterranean	Jan. 2021	NDFI/1.0 DII/1.0	67 % 75 %	3,293	3.41	0.014	1.52	Study Area
Kishima, Saga, Japan	Subtropical	Aug. 2021	NDFI/0.8 DII/0.8	–– ––	2,161	9.77	0.392	7.87	Study Area
Florida, USA	Subtropical	Sep. 2022	NDFI/1.4 DII/1.4	67 % 66 %	29	19.1	0.059	0.7	App's Report
Marracuene, Mozambique	Tropical	Feb. 2023	NDFI/1.5 DII/1.5	–– ––	3,403	113.06	0.012	32.83	App's Report
Wilmington, NC, USA	Subtropical	Aug. 2018	NDFI/1.0 DII/0.8	52 % 54 %	24	7.55	0.402	0.03	App's Report
Huoqiu County, Anhui, China	Subtropical	Aug. 2020	NDFI/1.5 DII/1.5	–– ––	163,010	678.02	1.707	609.47	App's Report

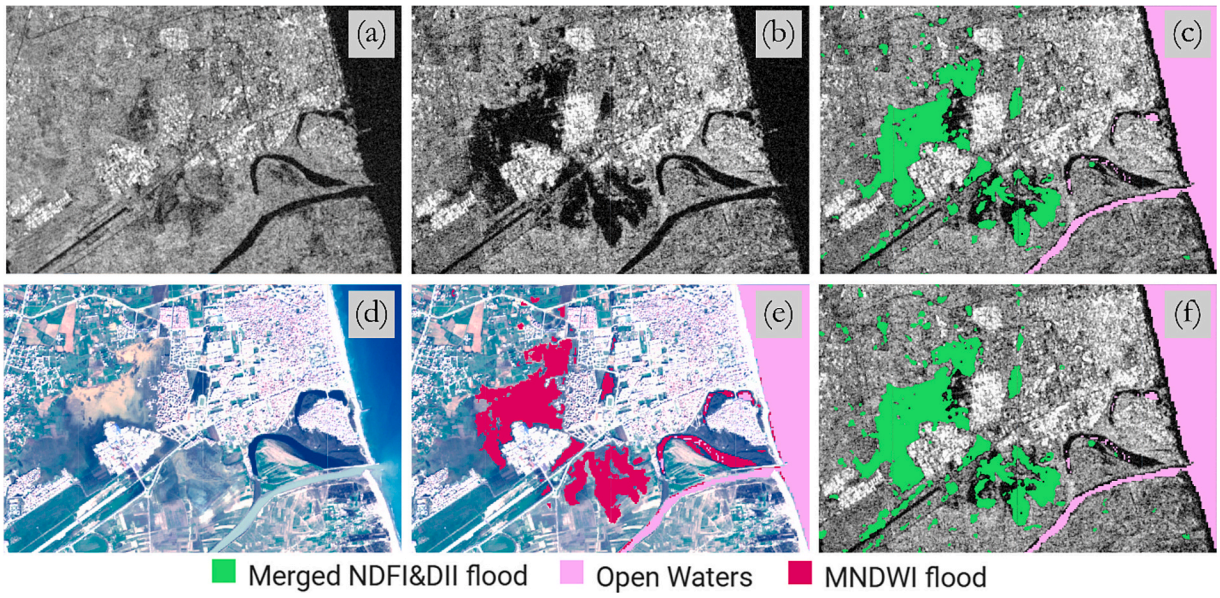


Fig. 6. Tetouan flood: SAR images (a) before and (b) after the flood and (c) extracted flood areas using the predefined $k_f = 1.5$ (green). (d) Post-flood optical image and (e) extracted flood areas (red). (f) SAR flood map using optimal k_f values calculated from sensitivity analysis ($k_f = 1.0$; refer to Fig. 7). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

make informed decisions for both present and future aid.

Additional flood mapping examples are provided in the App manual report (Hamidi and Peter, 2024), and summarized in Table 3, demonstrating its ease of use and global applicability. The App manual report is also provided in the supplementary material.

4. Discussion

Floods are frequent and complex natural disasters with rising occurrence and severity, threatening human life and assets, which has prompted responsible authorities to prioritize effective flood management. Therefore, providing flood extent maps during and after events is crucial for flood impact mitigation. Accordingly, organizations like FEMA, NOAA, NASA, ECMWF, NSERC, GFP, and SERVIR are dedicated to flood risk management and developing monitoring tools. For example, FEMA's flood inundation maps are valuable for assessing flood risk; however, they rely primarily on flood probability models rather than observed data. This reliance overlooks the impacts of emergent climate change and is further constrained by infrequent updates, relatively coarse spatial resolution, and dependence on static scenarios. As a result, these maps are less effective in capturing recent land use changes or dynamic flood events. Moreover, GloFAS provides early warning of flood events by integrating ECMWF weather forecasts with the LISFLOOD hydrological model. GloFAS provides global coverage, includes uncertainty estimates from ensemble forecasts, and offers open-access data, making it especially useful in regions with limited data availability. However, its coarse spatial resolution (approximately 5 km) and reliance on modeled inputs limit its accuracy at local scales, particularly in urban or small catchments. Other tools recommended by NOAA and USGS, such as the Flood Inundation Mapping (FIM) platform, generate DEM-based flood maps that depend heavily on the accuracy of elevation data. These models require validation using observed gauge heights or remote sensing data. Our App can serve as a valuable and efficient complementary tool for rapidly validating FIM and GloFAS outputs using high resolution (10 m) remote sensing-derived flood extent maps, which are available in all-weather and day/night conditions.

The App requires only the flood event period and a user-defined

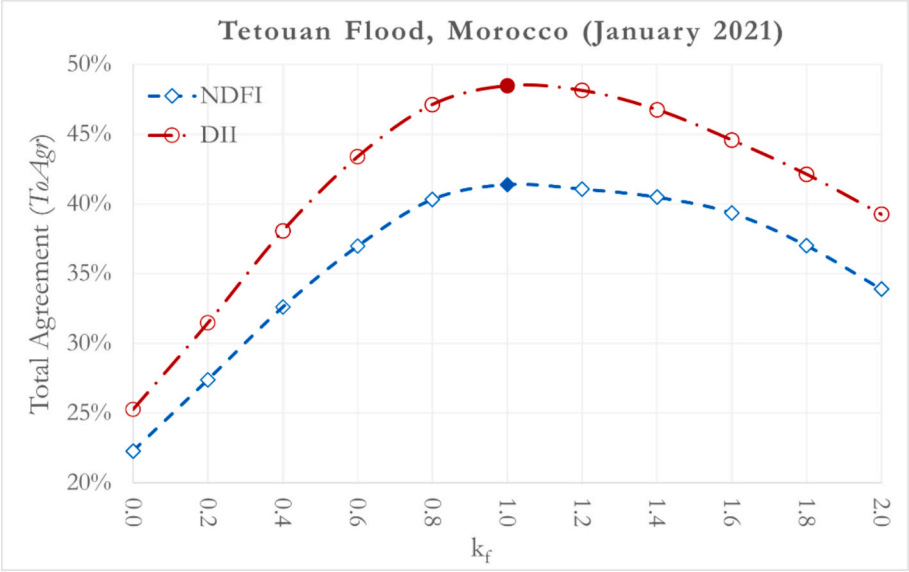


Fig. 7. Sensitivity analysis of NDFI and DII threshold values compared with optical flood inundation map to find optimum k_f values for Tetouan flood, January 2021.

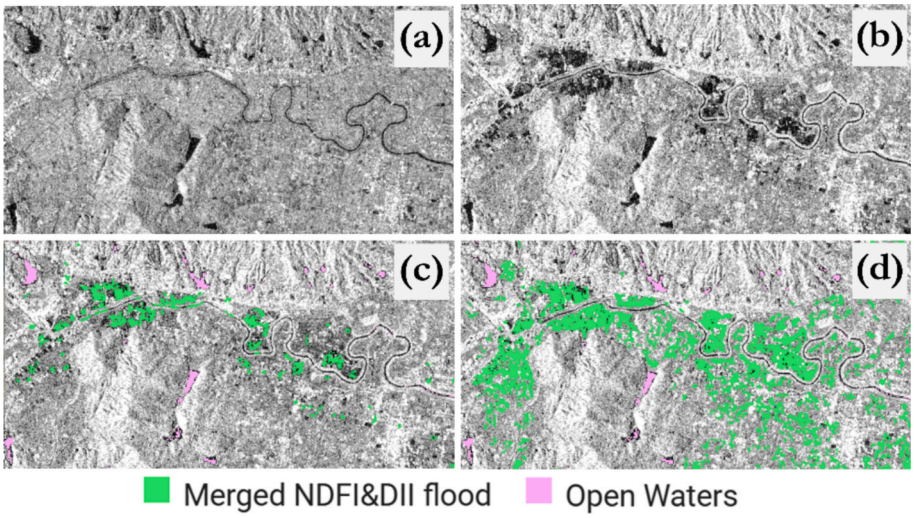


Fig. 8. Saga flood: SAR image (a) before, and (b) after the flood, (c) extracted flood areas using the predefined $k_f = 1.5$ (green), and (d) extracted flood areas using the user defined k_f value (green), comparing before and after flood SAR images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

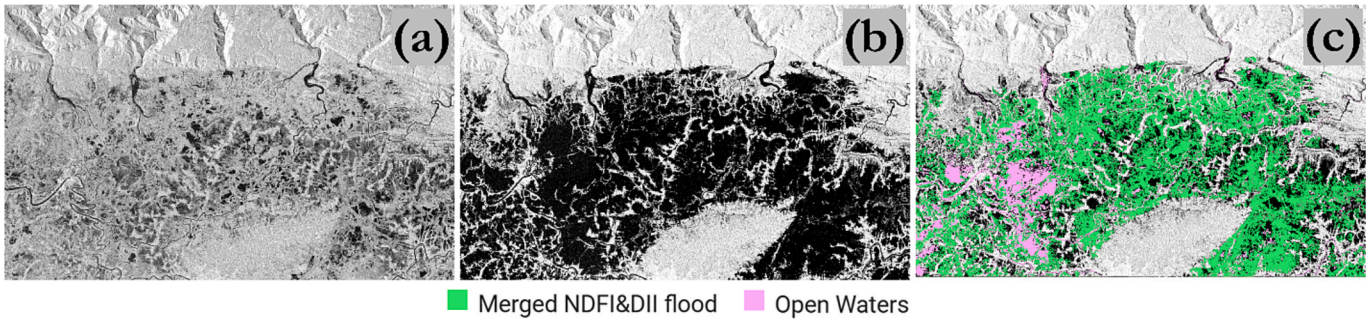


Fig. 9. Sylhet flood: SAR images (a) before, (b) after the flood, and (c) extracted flood areas using $k_f = 0.5$ (green), comparing before and after flood SAR images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

study area boundary to operate, making it fast and user-friendly. Via personal correspondence, the tool been successfully used by other researchers with limited experience in remote sensing to extract flood extents for analyzing land use and population impacts. The App leverages the GEE platform, eliminating the need for image downloading, preprocessing, post-processing, and manual flood map extraction. All computations are performed in the cloud using publicly available datasets, requiring no local computing resources. For instance, Sentinel-1 and Sentinel-2 data are accessible via the GEE cloud platform within approximately eight hours of data acquisition, making them suitable for rapid flood mapping. This benefits decision-makers and first responders by providing near-real-time information to effectively protect lives and property. The underlying JavaScript code is fully automated, and performance has been improved through code optimization to minimize user input and reduce processing time. For example, results were generated in 43 s for the Coffeyville flood event (section 3.1) and 35 s for the Sylhet flood event (section 3.4). However, processing time (e.g., for determining optimum SAR thresholds) may vary depending on the size of the study area. Additionally, the accuracy of the multidisciplinary information should be assessed independently.

To enhance the adaptability of the proposed tool for global flood mapping, threshold values are dynamically assigned based on the characteristics of the available imagery for each specific flood event and location. This approach supports the generalizability of the tool across diverse climatic conditions and land cover types worldwide. Furthermore, employing change detection indices derived from pre- and post-flood SAR imagery enhances flood delineation accuracy by reducing the impact of background surface texture, flood event characteristics, and environmental variability. This method effectively highlights flood-induced changes while minimizing irrelevant variations, making it resilient to climatic variations and independent of weather conditions. Table 3 specifies the climate conditions of each study area, demonstrating the effectiveness of SAR-based flood extraction across diverse regions.

The tool initially uses a predefined threshold value of $k_f = 1.5$, as suggested by previous studies (Cian et al., 2018; Clement et al., 2018; Long et al., 2014); however, this fixed value may not be appropriate for all SAR indices or case studies (Hamidi et al., 2023). To improve accuracy, the tool performs an automated sensitivity analysis using available optical flood maps, even if partially cloud-covered, to determine event- and location-specific thresholds (e.g., Coffeeville and Morocco case studies). When optical imagery is unusable due to full cloud cover or cloud shadows, the tool defaults to $k_f = 1.5$ for calculating SAR index thresholds and also provides an interface for manual threshold adjustment. Users can visually inspect pre- and post-flood SAR images to refine estimates (e.g., Japan and Bangladesh case studies). Although manually derived maps may under- or overestimate actual extents, they still offer valuable approximations during early response, especially when guided by expert judgment. For optical imagery, the tool adopts a default threshold value based on literature (Phiri et al., 2021; Ferral et al., 2019; Baig et al., 2013) and allows user adjustment within a 0–0.5 range. A threshold of 0.4 is effective for cloud-free images, while lower values may improve flood detection in the presence of cloud cover or sun glint. While perfectly accurate reference data is unavailable, reliable information remains essential for validating remote sensing-based flood maps. Our approach employs optical Sentinel-2 imagery to improve SAR-derived flood extents, given its capability for accurate and straightforward flood detection. Despite the inherent limitations of optical data, our method provides a practical and efficient solution for rapidly calibrating SAR threshold values, particularly in regions where ground-truth data are lacking.

This App has some limitations, including temporal mismatches between SAR and optical acquisitions, and differences in spatial resolution between the datasets may introduce uncertainties in the final results. Discrepancies between optical and SAR data can lead to errors of commission, where the flood area is overestimated; conversely, if only single-source data are available, errors of omission may be common. Furthermore, flood mapping using optical imagery is prone to various sources of error, including atmospheric conditions, solar illumination, soil moisture, vegetation phenology, and the misclassification of cloud shadows as water, as well as the inability to detect flooding beneath vegetation, canopies, and debris. Moreover, the thresholding technique used for floodwater delineation using MNDWI may lead to under- or overestimation of flood extents, a limitation inherent in all threshold-based methods.

While SAR imagery provides valuable advantages for flood mapping, inherent limitations remain that may affect its performance under certain conditions. For instance, storm-induced surface roughness, resulting from wind, heavy rainfall, or wave activity, can create pseudo-roughness effects in SAR imagery, further complicating the identification of inundated areas. Furthermore, the SAR-based change detection method (Equations (1) and (2) identifies flooded areas by comparing pre- and post-event imagery. It effectively excludes permanent water bodies due to their consistently low backscatter (e.g., Fig. 6). However, its accuracy depends on the pre-event image accurately reflecting non-flood conditions; if it contains pre-existing water or misclassified surfaces, true flood areas may be missed. Moreover, for flood mapping in under-vegetation and densely developed urban areas, radar shadows, limited signal penetration, and signal attenuation may lead to misclassification and make it difficult to detect flood water bodies. Some of these limitations arise from the use of C-band SAR products from Sentinel-1, which offer limited penetration through vegetation and built-up areas and are better suited for flood mapping in sparsely vegetated, rural and moderately developed regions. For enhanced flood detection in more complex environments, alternative SAR data offers distinct advantages. L-band SAR products, such as ALOS-2, SAOCOM, and the upcoming NASA-ISRO NISAR, are well-suited for detecting inundation under vegetation due to their longer wavelengths and ability to penetrate forest canopies, utilizing double-bounce scattering to identify sub-canopy flooding, particularly in wetlands and forested floodplains. In contrast, X-band SAR systems, such as the high-resolution, rapid-revisit Capella satellite products, are ideal for urban flood mapping. Their fine spatial resolution (up to 0.3 m) enables detailed detection of floodwaters in densely built environments, including streets, infrastructure, and narrow water bodies.

In this study, we utilized VH polarization from Sentinel-1 products for flood inundation mapping, which has also been reported by Marghany (2022b) and other researchers as a suitable SAR polarization for surface water detection (Kuntla and Manjusree, 2020; Tripathi et al., 2020; Manavalan et al., 2017; Manjusree et al., 2012). However, other SAR polarizations have been used for various applications. For instance, Manjusree et al. (2012) demonstrated that HH polarization offers clearer land–water separation, while VV polarization is effective in identifying partially submerged features, which is valuable for flood damage assessment. HH has also been widely applied for sea ice monitoring due to its sensitivity to structural features, whereas VH is responsive to vegetation and surface roughness, aiding in land cover classification (Marghany, 2022b). In agricultural applications, Veloso et al. (2017) highlighted the potential of SAR polarizations for crop type classification, phenological stage detection, and estimation of biomass and green area index. Their findings indicate that VH and VV are effective for distinguishing winter crops, while the VH/VV ratio is valuable for identifying summer crops and retrieving biophysical parameters such as crop biomass and irrigation status.

5. Conclusion

Satellite-based products are crucial for managing floods and meeting diverse needs such as monitoring, risk mitigation, and emergency planning. This paper introduces a user-friendly App using multi-source remote sensing data for global rapid flood monitoring. It empowers decision-makers, first responders, scientists, and stakeholders with vital

information to effectively understand and respond to floods, protecting at-risk populations and infrastructure.

This study uniquely combines optical and SAR-based flood detection methods, geospatial considerations (e.g., the modifiable areal unit problem), and automated thresholding within a cloud-based environment. The tool's speed supports rapid flood extent mapping, and its design requires minimal user interaction, enabling reliable and scalable outputs. Furthermore, the App is open source, promoting transparency, reproducibility, and opportunities for future enhancement.

In future studies, this App may support the calibration of numerical flood models, as its outputs are easily downloadable and can be directly compared with other types of flood inundation maps. Integrating remote sensing flood maps with numerical modeling outputs can enhance flood inundation mapping by combining their strengths to produce daily or hourly flood estimates, addressing the limitations of remote sensing's low temporal resolution that may overlook rapid flood dynamics. Subsequent development of the App could include flood depth estimation to improve impact assessments and support more effective post-flood planning. While the current study leverages multi-source remote sensing imagery, digital elevation models, LCLU data, and other relevant datasets for flood mapping, incorporating data fusion techniques and machine learning algorithms could further improve result accuracy and expand the tool's applicability. Moreover, although this App currently relies on optical flood inundation maps to calibrate SAR threshold values, incorporating more accurate benchmarks, such as ground truth observations, would enhance the robustness of the validation process.

We continuously monitor newly available data sources added to the GEE cloud platform to ensure the App utilizes the highest-quality datasets. Additionally, we will regularly check the App for potential errors and address any issues reported by users through communication with the corresponding author, thereby supporting the App's long-term applicability and reliability. Additionally, the App's report is available in the Supplementary materials. The code, App, and report are available online at:

https://github.com/ebrahimhamidi/FastFloodMonitoringTool-FFMT.

https://turnkey-aleph-386916.projects.earthengine.app/view/geefastfloodmonitoring.

https://www.hydroshare.org/resource/bf66a6cc204d4691abda18833bf68760.

CRediT authorship contribution statement

Ebrahim Hamidi: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization, Project administration. **Brad G. Peter:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization, Project administration. **Hamed Moftakhari:** Writing – review & editing, Funding acquisition, Project administration. **Hamid Moradkhani:** Writing – review & editing, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hamid Moradkhani reports financial support was provided by US Army Corps of Engineers. Hamed Moftakhari reports financial support was provided by US Army Corps of Engineers. Hamid Moradkhani reports financial support was provided by National Oceanic and Atmospheric Administration. Hamed Moftakhari reports financial support was provided by National Oceanic and Atmospheric Administration. Brad G. Peter reports financial support was provided by Consortium of Universities for the Advancement of Hydrologic Science, Inc. Ebrahim Hamidi reports financial support was provided by Consortium of Universities for the Advancement of Hydrologic Science, Inc. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jag.2025.104701.

Data availability

Data will be made available on request.

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